Kartekeyaan Raghavan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.34
Pace Junior Science College	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education)	2019	70.92%
Vivek Vidyalaya	Class X (Maharashtra State Board of Secondary and Higher Secondary Education)	2017	89.60%

WORK EXPERIENCE

LTIMindtree (LTM) | Cloud Engineer / Cybersecurity Consultant

- Led a **12-member SOC shift**, triaging **15 MDE, MDI and MDO incidents per shift** with **zero missed true positives** and containing **ransomware and phishing outbreaks** within **3-minute MTTA** and **30-minute MTTR** SLAs.
- Stopped active phishing campaigns through **email header, URL reputation and IOC analysis**, executing mailbox-wide purges that **neutralized credential harvesting**, and hunted lateral movement with **ATT&CK-aligned KQL**, pivoting on **IOCs**, **threat intelligence** and **Windows Event IDs.**
- Engineered **10+ custom KQL advanced-hunting detections** for a **multi-tenant Defender XDR** estate and escalated **false-negative gaps** found during live triage to Microsoft engineering, mapped to **MITRE ATT&CK**.
- Tuned **15 high-volume detection rules**, improving the product's **Signal-to-Noise Ratio (SNR)** and cutting **false positives by 45%**, and spearheaded AI-STM automation whose **auto-resolution workflows** cut investigation time by **33%**
- Developed **investigation playbooks** and optimized **SOPs**, delivered **root-cause analyses** and **Grafana-backed reviews** to **senior Microsoft stakeholders**, and raised **P0/P1 product bugs** with engineering.

CERTIFICATIONS & ACHIEVEMENTS

- **CEH** v13: **Certified Ethical Hacker**, EC-Council
- **SC-200** Microsoft Certified Security Operations Analyst Associate
- **Top 4 nationally (team lead)**, Deutsche Börse Cyber Security Challenge, IIT Hyderabad (2026)
- **Won national level Techxter** 12.0 National Technical Paper Presentation at SIES GST
- **Super Crew** and **Hi-Five** spot awards for teamwork and leadership, LTM (2024-2025)

PROJECTS

Autonomous Web-Exploitation Agent on Local SLMs (M.Tech Thesis) | GitHub

- Benchmarked models **(1.5B-32B)** across **600+** attack runs on **3** vulnerable web apps, architecting a **knowledge distillation** pipeline from a massive **235B** teacher down to a local **SLM**.
- Pioneered a novel **failure-scoring metric (WDFM)** exposing critical AI gaps like a **32B** model that successfully landed **96%** of exploits but accurately recognized its own success state only **44%** of the time.
- Quantified **Defensive Refusal Bias** proving models incorrectly refused authorized security tasks **27%** more often, and recovered this capability via **task-preserved knowledge distillation** on **agentic trajectories**.

TransactiWar: Hardened Banking Web Application | GitHub

- Built and strengthened a **banking web app** with **88 security defenses,** spanning from secure password hashing to **tamper-proof money transfers**, successfully resolving **17 vulnerabilities** during a live attack-and-defend war game.
- Secured money transfers using a **2-phase** commit system with **HMAC integrity tokens**, mitigating **race conditions** via integer-cents arithmetic and database-level ordered row locking.
- Defeated **brute-force attacks** via **dual-layer rate limiting** and **progressive server delays**, alongside custom database-backed session management and secure **Argon2id** password hashing.

DNSSEC Chain-of-Trust Validator | GitHub

- Engineered a **custom DNS resolver** from scratch that rigorously validates the **DNSSEC** cryptographic **chain of trust**, verifying signatures from internet root servers down to the target domain.
- Architected a **resilient** manual **cryptographic fallback pipeline** that directly parses **RRSIG** bytes and **DNSKEY** algorithms if standard validation libraries fail or are unavailable.

Collision-Aware Motion Prediction for Autonomous Vehicles | Github

- Optimized a state-of-the-art **deep learning motion-prediction architecture** for **autonomous vehicles**, engineering enhancements that successfully **surpassed** the original academic paper's baseline accuracy.
- Engineered a **geometric consensus selection method** to replace flawed, near-uniform confidence scores, driving a **6.1%** improvement in **top-1 trajectory** prediction accuracy.

SKILLS

- **SOC Operations:** Alert triage, incident response, threat hunting, phishing and IOC analysis, vulnerability management
- **Platforms:** Microsoft Defender XDR (MDE, MDI, MDO, MCAS), Wazuh SIEM/XDR, EDR, Grafana
- **Languages and Tools:** Kusto Query Language (KQL), Sigma, Python, C++, Bash, Git, Nmap, Wireshark, Burp Suite
- **Frameworks:** MITRE ATT&CK, Cyber Kill Chain, Detection-as-Code, OWASP Top 10, STRIDE
- **Networking and Systems:** Network security, Linux, TLS and PKI, DNSSEC
- **Soft Skills:** Problem Solving, Leadership, Analytical Thinking, Cross-Functional Collaboration, Ownership & Accountability
- **Languages:** English, Marathi, Tamil, Hindi

POSITIONS OF RESPONSIBILITY

- **Core Member** of the **IITH SOC Team**, triaging alerts from an **institute-wide Wazuh deployment**
- **Core Member** of **DRISHTE**, IIT Hyderabad's security, threats and investigation research lab

- **Teaching Assistant** for **Network and Information Security**, mentoring 10 sophomore students